

USENIX

Overall Review

USENIX Security Symposium: Bridging Theory and Practice in Adversarial Machine Learning (2022-2025)

Research at the USENIX Security Symposium from 2022 through 2025 reveals a persistent but evolving tension between theoretically rigorous adversarial machine learning methodologies and the practical constraints of real-world deployment. While the conference has historically served as a venue for advancing both attack sophistication and defensive mechanisms, a consistent pattern emerges: many proposed solutions fail to address the fundamental trade-offs between security, utility, and computational efficiency that govern industrial adoption decisions.

The Utility-Robustness Trade-off

The most significant barrier to deployment remains the conflict between achieving robust security guarantees and maintaining acceptable system performance. Early work identified this challenge across multiple domains: defensive proposals rooted in established academic practices often achieved certifiable robustness but at the cost of poor clean classification accuracy, which "discourages the real-world deployment" of these defenses (Xiang et al., 2022). This pattern persisted through subsequent years, with certifiably robust defenses in computer vision requiring 10 to 100 times more inference-time computation than undefended models, rendering them computationally prohibitive for practical use (Xiang et al., 2024). The problem extends beyond vision systems—defenses against malicious activations in voice assistants typically employ generic machine learning methods that "harm the natural accuracy—an unacceptable proposition" for commercial systems (Ahmed et al., 2022). Similarly, widely used privacy-preserving techniques like DP-SGD compromise model utility significantly to achieve privacy guarantees (Liu et al., 2022; Tang et al., 2022).

However, more recent research demonstrates movement toward resolving this fundamental conflict. PatchCleanser introduced a double-masking approach compatible with any image classifier, achieving high certified robustness while preserving state-of-the-art clean accuracy (Xiang et al., 2022). EKOS circumvented the utility trade-off inherent in generic defenses by leveraging the physical and semantic properties of keyword spotting tasks (Ahmed et al., 2022). By 2024, MIST offered a pathway toward robust security without performance penalties by strategically limiting overfitting only to the most membership-vulnerable training instances (Li et al., 2024). The 2025 introduction of CAMP training demonstrated that provable adversarial robustness in deep reinforcement learning need not sacrifice certified expected return—crucial for safety-critical robotics applications (Wang et al., 2025).

Misalignment Between Academic Threat Models and Industrial Realities

A critical meta-analysis exposed widespread methodological pitfalls in security research, including reliance on "lab-only evaluation" and deployment of "inappropriate threat models" that fail to account for adaptive adversaries (Arp et al., 2022). This critique gained empirical support as research progressed: academic studies often operate under assumptions of overly generous attacker access—such as extensive access to internal models, parameters, or training data—that do not reflect the stringent security controls present in real-world corporate environments (Grosse et al., 2024). As practical observations emphasized, real attackers don't compute gradients; they exploit simpler, more accessible vulnerabilities that academic research often overlooks in favor of sophisticated but impractical attack methods.

This misalignment manifests across application domains. In resource-constrained environments like on-device machine learning for IoT, proposed academic attacks and defenses prove difficult to reproduce, fail to perform effectively on production models, or introduce unacceptable computation and energy costs (Nayan et al., 2024). Evaluation methodology itself suffers from these issues—deepfake detection research uses inappropriate performance metrics and unrealistic dataset distributions, leading to overestimation of detector efficacy and creating difficulties for transitioning these tools into practice (Layton et al., 2024). Even the

presumed capability of machine learning for automatic vulnerability detection has been challenged, with state-of-the-art models largely overfitting to unrelated features in training data, rendering them incapable of generalizing to slightly modified scenarios and providing a "false sense of security" (Risse & Böhme, 2024).

Evolution of Practical Attack Vectors

Despite these methodological concerns, research has progressively shifted toward practical attack vectors that address real-world constraints and deployment scenarios. Early work demonstrated physically realizable attacks such as the "frustum attack," which leveraged environmental context to compromise automotive sensor fusion systems in black-box settings (Hallyburton et al., 2022). This practical orientation intensified in 2023 with attacks designed for operational constraints: HARDBEAT generated high-success-rate triggers needing only knowledge of final predicted labels and minimal queries, addressing restrictions imposed by proprietary commercial services (Tao et al., 2023). Physical and analog attack surfaces received increased attention, including physically realizable 3D adversarial objects optimized for shape to deceive X-ray prohibited item detection (Liu et al., 2023), focused laser spoofing to selectively remove LiDAR point cloud data (Cao et al., 2023), and analog adversarial attacks where human adversaries manipulated voice signals using simple tubes to bypass speaker recognition (Ahmed et al., 2023).

By 2024, attacks began capitalizing on simplicity and accessibility. Jailbreaking large language models proved highly effective even when executed by inexperienced users via strategically crafted natural language prompts, emphasizing the potency of low-cost, accessible black-box attacks over complex gradient optimization (Liu et al., 2024; Yu et al., 2024). The 2025 research landscape further embraced practical deployment scenarios: "Shadow Hack" exploited LiDAR weaknesses using ordinary non-reflective materials on roads (Kobayashi et al., 2025), while ATKSCOPES demonstrated rapid evasion against real-world perceptual hashing algorithms by dynamically adapting perturbation scale to victim algorithms' feature extraction mechanisms (Zhang et al., 2025). Attacks also prioritized stealth and persistence—MergeBackdoor revealed supply chain threats where seemingly benign upstream models activate malicious backdoors upon merging (Wang et al., 2025), while persistent backdoor strategies

targeting stable neuronal components ensured exploits survive continuous parameter updates in continual learning systems (Guo et al., 2025).

Specialized and Context-Aware Defenses

Defensive research has evolved from generic solutions toward specialized, interpretable, and context-aware mechanisms. Blacklight successfully detected and mitigated nearly all black-box query-based attacks against MLaaS by leveraging the fact that iterative optimization inevitably produces highly similar queries, providing effective defense against persistent attackers that bypass account-based security measures (Li et al., 2022). The Universal Robustness Evaluation Toolkit (URET) addressed the challenge of generating adversarial inputs for modalities beyond typical image classification by formulating adversarial generation as a graph exploration problem, seeking sequences of domain-specific, functionality-preserving transformations for systems utilizing varied input objects like malware binaries or tabular data (Eykholt et al., 2023).

Recent defenses demonstrate increased sophistication and specialization. JBSHield moved beyond heuristics for large language model protection by using the Linear Representation Hypothesis to identify and manipulate "toxic" and "jailbreak" concepts within hidden states (Zhang et al., 2025). SafeSpeech proactively poisoned voice data during training to make synthesized audio low-quality and unusable, achieving robustness surpassing inference-only defenses against voice cloning (Zhang et al., 2025). CertPHasH introduced the first certified perceptual hashing system with robust training verified against evasion and collision attacks (Yang et al., 2025). Notably, DeBackdoor addressed realistic deployment constraints by providing backdoor detection effective under black-box access, data scarcity, and pre-deployment inspection limitations (Popovic et al., 2025).

System-Level Vulnerabilities and Emerging Threats

Beyond individual model vulnerabilities, research increasingly recognizes that integrating machine learning models into complex, multi-component systems fundamentally undermines traditional security and privacy assumptions. While prior work focused on mitigating leakage from isolated models, system-level components such as training data filters or output monitoring introduce critical

privacy side channels easily exploitable by adaptive adversaries, often invalidating provable differential privacy guarantees (Debenedetti et al., 2024). Application-specific attacks like CDMI successfully targeted industry-adopted document understanding models, demonstrating that security research struggles to keep pace with deployment of powerful, commercially viable multimodal architectures (Dentan et al., 2024). CleanSheet revealed that model hijacking achieving high attack success rates requires only substantially weaker, more practical adversary assumptions—partial knowledge of clean training data—underscoring vulnerabilities inherent in real-world model deployments relying on public datasets (Ge et al., 2024).

Emerging architectures present novel challenges to outdated defenses. The personalization component of Personalized Federated Learning, while designed to enhance utility, introduces bottlenecks that traditional backdoor attacks overcome by aligning backdoor gradient and loss objectives with the main task, thereby evading both server-end and client-end defense mechanisms designed for non-personalized settings (Lyu et al., 2024). Privacy-focused attacks increasingly target upstream processes, as demonstrated by data poisoning attacks against unlabeled pre-training data used in single-modal contrastive learning (Liu et al., 2022). Even ostensibly benign engineering choices introduce vulnerabilities—gradient compression techniques intended to optimize communication in Federated Learning environments inadvertently create a "false sense of security" against privacy-stealing attacks, shifting focus away from comprehensive defense mechanisms (Yue et al., 2023).

Organizational and Systemic Barriers to Adoption

Beyond technical challenges, qualitative research reveals significant organizational and educational barriers preventing industry adoption of adversarial machine learning defenses. Machine learning practitioners often lack institutional motivation and usable resources to understand and mitigate adversarial threats, frequently assuming security and machine learning are entirely disconnected fields (Mink et al., 2023). This detachment results in low prioritization of adversarial evaluations before deployment and poor visibility into monitoring for active attacks. Consequently, defenses remain unimplemented due to isolation between ML and security teams or because competing business

priorities outweigh the cost and time needed for robust implementation (Mink et al., 2023).

Addressing these systemic issues requires focusing on fundamental engineering challenges. RDMA acceleration for secure collaborative machine learning protocols addresses performance barriers that preclude real-world deployment (Ren et al., 2024). PatchCure demonstrates that improving both certifiable robustness and computational efficiency simultaneously is achievable through careful system design (Xiang et al., 2024). These efforts suggest that bridging the research-practice gap necessitates improvements not only in creating robust, realistic defensive technology but also in fostering organizational structures and resources that prioritize adoption of critical security measures.

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Year Breakdown

1.) 2022

The ongoing challenge in adversarial machine learning research lies in bridging the gap between theoretically rigorous methodologies and practical real-world deployment constraints, particularly concerning model utility, security, and

efficiency. Several research efforts highlight this tension, noting that defensive proposals rooted in established academic practices often fail against realistic threats or impose unacceptable operational costs. For instance, while prior works achieved *certifiable robustness* against adversarial patches, their dependence on specific model architectures frequently yielded poor clean classification accuracy, which inherently "discourages the real-world deployment" of these defenses (Xiang et al., 2022). Similarly, existing defenses against highly practical threats like malicious activations in voice assistants rely on generic machine learning methods that "typically harm the natural accuracy—an unacceptable proposition" for commercial systems (Ahmed et al., 2022). A holistic assessment of ML systems confirms that widely used defenses such as DP-SGD often compromise model utility significantly to achieve privacy guarantees (Liu et al., 2022; Tang et al., 2022), further revealing this fundamental trade-off. A critical meta-analysis suggests that this persistent gap is fueled by widespread methodological pitfalls in security research, such as relying on "lab-only evaluation" and failing to deploy an "inappropriate threat model" that accounts for adaptive adversaries (Arp et al., 2022).

Recent research attempts to shift focus towards practical robustness, resulting in both more subtle attacks and more robust, high-utility defenses. The discovery of novel vulnerabilities underscores the severity of the threat landscape: attackers can now leverage subtle properties to bypass defenses in complex deployment scenarios, such as the physically realizable "frustum attack," which uses environmental context to compromise automotive sensor fusion systems in a black-box setting (Hallyburton et al., 2022). Additionally, privacy-focused attacks are targeting upstream processes, as demonstrated by data poisoning attacks against the unlabeled pre-training data used in single-modal contrastive learning, a vector where conventional defenses are ineffective (Liu et al., 2022).

Conversely, defenses like PatchCleanser demonstrate success in addressing the utility-robustness conflict by using a double-masking approach that is compatible with *any* image classifier, thus achieving high certified robustness while preserving state-of-the-art clean accuracy (Xiang et al., 2022). For Keyword Spotting, EKOS achieves high precision and security by leveraging the physical and semantic properties of the task, circumventing the utility trade-off inherent in generic ML defenses (Ahmed et al., 2022). Addressing MLaaS threats, novel systems like Blacklight focus on black-box attacks, successfully detecting and

mitigating nearly all queries by leveraging the fact that iterative optimization inevitably produces highly similar queries, providing an effective defense against persistent attackers that bypass account-based security measures (Li et al., 2022). This movement towards defenses that prioritize both empirical security and high utility offers a pathway toward enhancing the robustness and trustworthiness of real-world deployed machine learning systems.

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2.) 2023

The evolving landscape of adversarial machine learning (AML) research is increasingly characterised by efforts to transition from theoretical, purely gradient-based perturbations to practical attacks that address real-world constraints, though significant gaps between academic insights and industry deployment persist. One major area of exploration involves generating adversarial inputs for modalities beyond typical image classification where models rely on inputs with complex, non-differentiable features, rendering traditional gradient computation infeasible (Eykholt et al., 2023). The **Universal Robustness Evaluation Toolkit (URET)** addresses this gap by formulating adversarial generation as a graph exploration problem, seeking sequences of domain-specific, functionality-preserving transformations rather than relying on differentiable feature spaces. This framework facilitates studying systems utilizing varied input objects like malware binaries or tabular data where semantic and functional correctness must be maintained during perturbation (Eykholt et al., 2023).

Furthermore, research is actively exploring physical and analog attack surfaces in safety-critical domains. For instance, the **X-Adv** approach designs physically realizable 3D adversarial objects optimized for shape—not color or texture—capable of deceiving X-ray prohibited item detection systems, highlighting the necessity of tailoring attacks to specific physical imaging principles and overcoming issues like complex object overlap in luggage (Liu et al., 2023). Similarly, **Physical Removal Attacks (PRA)** demonstrate the feasibility of using focused laser spoofing to selectively remove LiDAR point cloud data on autonomous vehicles, exploiting inherent sensor processing mechanisms to hide objects and induce dangerous driving decisions (Cao et al., 2023). Moving beyond digital perturbations entirely, the **"Tubes Among Us"** research successfully

demonstrated an **analog adversarial attack** where human adversaries manipulate their voice signal physically using simple tubes to misidentify speakers by voice recognition algorithms, effectively bypassing established digital artifact detection methods (Ahmed et al., 2023). The practicality challenge is also met through attacks designed specifically for operational constraints, such as the **Hard-label Black-box Universal Adversarial Patch Attack (HARDBEAT)**, which generates high-success-rate triggers needing knowledge only of the final predicted label (hard-label) and minimal queries, addressing restrictions often imposed by proprietary commercial services (Tao et al., 2023).

Despite these technical strides towards realistic attack vectors, the industry remains inadequately protected due to significant organizational and educational barriers. Qualitative root cause analysis indicates that machine learning practitioners often lack institutional motivation and usable resources to understand and mitigate **AML** threats, frequently assuming security and machine learning are entirely disconnected fields (Mink et al., 2023). This detachment results in a low prioritization of adversarial evaluations before deployment and poor visibility into monitoring for active attacks (Mink et al., 2023). Consequently, defenses remain unimplemented, either due to isolation between ML and security teams or because competing business priorities outweigh the cost/time needed for robust implementation (Mink et al., 2023). Furthermore, even ostensibly benign research or engineering choices can introduce vulnerabilities; for example, gradient compression techniques intended to optimize communication in Federated Learning environments can inadvertently create a "false sense of security" against privacy-stealing attacks like data reconstruction, shifting focus away from comprehensive defense mechanisms (Yue et al., 2023). This body of work collectively reveals that the research-practice gap is multi-faceted, requiring improvements not only in creating robust, realistic defensive technology, but also in fostering organizational structures and resources that prioritize the adoption of such critical security measures.

References

Citation	Passage Title
Ahmed et al., 2023	Tubes Among Us: Analog Attack on Automatic Speaker Identification

Bai et al., 2023	VILLAIN: Backdoor Attacks Against Vertical Split Learning
Cao et al., 2023	You Can't See Me: Physical Removal Attacks on LiDAR-based Autonomous Vehicles Driving Frameworks
Eykholt et al., 2023	URET: Universal Robustness Evaluation Toolkit (for Evasion)
Liu et al., 2023	X -Adv: Physical Adversarial Object Attacks against X-ray Prohibited Item Detection
Mink et al., 2023	"Security is not my field, I'm a stats guy": A Qualitative Root Cause Analysis of Barriers to Adversarial Machine Learning Defenses in Industry
Tao et al., 2023	Hard-label Black-box Universal Adversarial Patch Attack
Yue et al., 2023	Gradient Obfuscation Gives a False Sense of Security in Federated Learning
Zhang et al., 2023	CAPatch: Physical Adversarial Patch against Image Captioning Systems
Zhu et al., 2023	TPatch: A Triggered Physical Adversarial Patch

3.) 2024

The gap between adversarial machine learning (AdML) research and its practical deployment remains a persistent challenge, largely stemming from discrepancies between academic threat models and industrial realities (Grosse et al., 2024). Research often operates under assumptions of overly generous access—such as extensive access to internal models, parameters, or training data—that do not reflect the stringent security controls present in real-world corporate and service environments (Grosse et al., 2024). As emphasized by practical observations, real attackers don't compute gradients—they exploit simpler, more accessible vulnerabilities that academic research often overlooks in favor of sophisticated but impractical attack methods. This disparity means that many highly complex, gradient-based attacks proposed in academia may be impractical when targeting black-box systems. Furthermore, research into certifiably robust defenses in computer vision highlights an intrinsic trade-off: achieving strong robustness

often introduces massive computational overhead, potentially requiring 10 to 100 times more inference-time computation than undefended models, thus rendering these robust solutions computationally prohibitive for real-world use (Xiang et al., 2024).

This tension between theoretical complexity and real-world viability is evident across various ML application domains. In resource-constrained environments like on-device machine learning (ML) for IoT, researchers find that many proposed academic attacks and defences are difficult to reproduce, fail to perform effectively on production models, or introduce unacceptable computation and energy costs, thereby restricting their wider adoption (Nayan et al., 2024). Conversely, practical threats often capitalize on simplicity; for instance, jailbreaking large language models (LLMs) can be highly effective even when executed by inexperienced users via strategically crafted natural language prompts, emphasizing the potency of low-cost, accessible black-box attacks over complex gradient optimization (Liu et al., 2024; Yu et al., 2024). Moreover, shortcomings exist in evaluation methodology itself, particularly in domains like deepfake detection, where using inappropriate performance metrics (like EER) and unrealistic dataset distributions (such as overwhelming class imbalance) leads to an overestimation of detector efficacy, creating difficulties for transitioning these tools into practice (Layton et al., 2024). Ultimately, bridging this gap necessitates focusing on fundamental engineering challenges, such as improving the efficiency of protocols used in secure collaborative ML (SCML) via solutions like RDMA acceleration, thereby addressing performance barriers that preclude real-world deployment (Ren et al., 2024).

The inadequacy extends to system-level integration, where integrating ML models into complex, multi-component systems fundamentally undermines traditional security and privacy assumptions. While prior work focuses on mitigating leakage from isolated models, Debenedetti et al. (2024) reveal that system-level components, such as training data filters or output monitoring, introduce critical privacy side channels that are easily exploitable by adaptive adversaries, often invalidating provable differential privacy guarantees. This failure is mirrored in application-specific contexts, as Dentan et al. (2024) developed the CDMI attack specifically targeting industry-adopted document understanding models (LayoutLM and BROS), demonstrating successful reconstruction of sensitive data fields, indicating that security research struggles to keep pace with the

deployment of powerful, commercially viable multimodal ML architectures. Bridging the gap regarding realistic attack capabilities, Ge et al. (2024) introduce CleanSheet, a model hijacking technique that achieves high attack success rates (comparable to resource-intensive backdoor attacks) while relying on substantially weaker, more practical adversary assumptions—namely, only partial knowledge of the clean training data—underscoring a crucial vulnerability inherent in real-world model deployments that often rely on public datasets.

The inadequacy of current ML security techniques is further evidenced by failures in foundational tasks and emerging architectures, exposing a fragility in models trained specifically for security objectives. Risse and Böhme (2024) challenge the presumed capability of machine learning for automatic vulnerability detection (ML4VD), finding that existing state-of-the-art models largely overfit to unrelated features in training data, rendering them incapable of generalizing to even slightly modified scenarios, such as distinguishing vulnerable code from its patched counterpart, suggesting that the high accuracy scores reported on standard benchmarks often provide a "false sense of security". Furthermore, emerging federated learning systems present novel challenges to outdated defenses; Lyu et al. (2024) show that the personalization component of Personalized Federated Learning (PFL), while designed to enhance utility, introduces bottlenecks that traditional backdoor attacks overcome by aligning backdoor gradient and loss objectives with the main task (PFedBA), thereby evading both server-end and client-end defense mechanisms designed for non-personalized settings. Conversely, research like Li et al. (2024) seeks to minimize the inherent utility degradation often introduced by conservative defenses by proposing MIST, which strategically limits overfitting only to the most membership-vulnerable training instances, offering a pathway toward robust security assurances without the accompanying performance penalty.

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4.) 2025

Recent adversarial machine learning (AML) research reveals a complex landscape shifting from purely theoretical attacks toward practical, context-specific threats and defenses. This shift directly addresses the realism gap identified in prior work. Novel physical-world attacks are emerging: *Kobayashi et al. (2025)* introduce "Shadow Hack," exploiting LiDAR weaknesses using ordinary non-reflective materials placed on roads to trigger false object detection in autonomous vehicles. *Zhang et al. (2025)* bridge the gap between academic and commercial attacks with **ATKSCOPES**, demonstrating rapid evasion and collision against real-world perceptual hashing algorithms by dynamically adapting perturbation scale to the victim algorithm's feature extraction mechanism. Attacks now prioritize stealth and persistence: *Wang et al. (2025)* reveal **MergeBackdoor**, a supply chain threat where seemingly benign upstream models pass security checks but activate malicious backdoors upon merging. *Guo et al. (2025)* propose persistent backdoor strategies that survive continuous parameter updates in **Continual Learning** by targeting stable neuronal components, ensuring exploits persist across tasks.

Defenses are becoming more specialized, interpretable, and certifiably robust. For Large Language Models, *Zhang et al. (2025)* introduce **JBSHIELD**, moving beyond heuristics by using the Linear Representation Hypothesis to identify and manipulate "toxic" and "jailbreak" concepts within hidden states. For **voice cloning**, *Zhang et al. (2025)* propose **SafeSpeech**, proactively poisoning voice data during training to make synthesized audio low-quality and unusable—achieving robustness that surpasses inference-only defenses. *Yang et al. (2025)* introduce **CertPHasH**, the first certified perceptual hashing system with robust training verified against evasion and collision attacks. However, deployment often involves strict resource constraints that academic research neglects. *Popovic et al. (2025)* address this with **DeBackdoor**, a backdoor detection framework effective under realistic limitations: black-box access, data scarcity, and pre-deployment inspection. Finally, *Wang et al. (2025)* introduce **CAMP** training to enhance certified expected return in Deep Reinforcement Learning agents without sacrificing provable adversarial robustness—crucial for safety-critical robotics.

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